

Symmetric Factorized Quadratic Adaptation: A Controlled Study of Nonlinear Low-Rank Adapters for GPT-2

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Abstract

We compare LoRA with an activation-dependent symmetric quadratic adapter at matched parameter counts in frozen GPT-2. An independent 24-run WikiText-2 confirmation favors Symmetric in mean final validation and held-out loss at ranks 1, 2, 4, and 8, whereas six controlled AG News runs favor LoRA in mean accuracy and macro-F1. A single-seed scaling ablation, convergence curves, local-Jacobian spectra, adapter Hessians, and a matched cost benchmark show that the result is task and protocol dependent rather than universal.

1 Introduction

Parameter-efficient fine-tuning adapts a pretrained model while keeping its backbone frozen. We ask whether a quadratic function of a projection input can offer useful capacity at the same parameter count as a LoRA branch. This is an independently evaluated formulation and controlled experimental study, not a claim that quadratic low-rank networks are unprecedented.

2 Related Work and Positioning

The methods below are compared at the mathematical and architectural levels. Published benchmark results obtained with different backbones, datasets, training budgets, or evaluation protocols are not numerically ranked against the experiments in this study.

LoRA [1] learns the low-rank linear correction $\frac{\alpha}{r}(xA)B$ and is the direct experimental baseline. DoRA [2] decomposes weight magnitude and direction. MoRA [3] uses a square trainable matrix with nonparametric transforms and is designed to increase effective update rank, while HiRA [4] uses a Hadamard-product high-rank weight update. LoRAN [5] applies a nonlinear transformation to the low-rank weight update. These are contextual weight- or parameter-space methods and were not experimentally reproduced here.

2.1 PERA

PERA [6] introduces polynomial expansion in parameter and low-rank-factor space before factor composition. This study instead applies the nonlinearity directly to projected activations:

$$x \longrightarrow xU \longrightarrow (xU) \odot (xU) \longrightarrow P.$$

Both approaches investigate higher-order structure in parameter-efficient adaptation, but the polynomial or quadratic operation acts on different mathematical objects. PERA was not used as an experimental baseline.

2.2 QuadraNet V2

For one output coordinate of the Symmetric adapter,

$$\Delta y_j = x Q_j x^T, \quad Q_j = \frac{\alpha}{r} U \text{diag}(P_{:,j}) U^T.$$

This places the method within the family of factorized quadratic transformations represented by work such as QuadraNet V2 [7]. Across a complete multi-output layer, all output coordinates share the projection directions contained in U , while P provides signed output-specific coefficients. Consequently, similarity at the single-output quadratic-form level does not establish complete architectural equivalence. QuadraNet V2 is treated as a closely related quadratic precedent, not evidence of either exact equivalence or universal novelty.

2.3 Position of the proposed method

The defining feature of the Symmetric adapter is that its low-rank projection is applied to the incoming activation and the projected coordinates are squared before being mapped back to output space:

$$\Delta y_{\text{Symmetric}} = \frac{\alpha}{r} ((xU) \odot (xU)) P.$$

This produces an activation-dependent quadratic correction with an input-dependent Jacobian while retaining a parameter structure comparable in size to the matched LoRA branch. The experiments evaluate this design directly against LoRA under controlled budgets. DoRA, MoRA, HiRA, LoRAN, PERA, and QuadraNet V2 clarify the literature context and are not reproduced baselines.

3 Method

The adapter is trained beside a frozen GPT-2 projection rather than replacing the pretrained model. Only the low-rank branch is optimized, allowing the functional form to be compared under matched trainable matrix shapes and parameter counts. For a row vector $x \in \mathbb{R}^{1 \times d}$, frozen $W \in \mathbb{R}^{d \times d_{out}}$, and frozen bias b , the base projection is $y_{base} = xW + b$. LoRA and Symmetric add

$$\Delta y_L = \frac{\alpha}{r} (xA)B, \tag{1}$$

$$\Delta y_S = \frac{\alpha}{r} ((xU) \odot (xU))P. \tag{2}$$

Here $A, U \in \mathbb{R}^{d \times r}$ and $B, P \in \mathbb{R}^{r \times d_{out}}$. Both methods contain $r(d + d_{out})$ trainable adapter parameters. For output j ,

$$\Delta y_{S,j} = x Q_j x^T, \quad Q_j = \frac{\alpha}{r} U \text{diag}(P_{:,j}) U^T, \tag{3}$$

so Q_j is symmetric and $\text{rank}(Q_j) \leq r$. The directions in U are shared across outputs; signed coefficients in P mean that Q_j need not be positive semidefinite. The expansion $(ax_1 + bx_2)^2 = a^2x_1^2 + 2abx_1x_2 + b^2x_2^2$ shows that the square creates pairwise cross terms. LoRA remains linear in its input, whereas Symmetric introduces explicit second-order interactions.

The controlled rank study uses $\alpha = 4$, so α/r equals 4, 2, 1, and 0.5 at ranks 1, 2, 4, and 8. The separate constant-scale ablation was motivated by this coupling of rank and effective scale.

4 Experiments

The work separates exploratory campaigns from controlled experiments. Early studies compared adapter families, training durations, attention and MLP placements, a full 12-block attention scan, and multilayer fixed-budget designs. Their protocols were not uniform, so they are not merged with controlled data.

Table 1: Completed experimental campaigns and their scientific roles.

Campaign	Design	Role
Adapter families	LoRA, quadratic, interaction, Symmetric, and mixed variants	Exploration
Depth/placement	Attention/MLP screens and 12-block scan	Placement evidence
Fixed-budget multi-layer	6,144 and 12,288 parameters across layers	Depth/local rank
AG News	Two methods, three seeds, 500 steps	Classification
Rank screening	Two methods, four ranks, three seeds, 1,000 steps	Screening
Rank confirmation	24 fresh runs, 5,000 steps	Central LM evidence
Scaling ablation	$\alpha = 4$ versus $\alpha/r = 1$, seed 42	Scale check
Mechanism/cost	Stored checkpoints and rank-4 benchmark	Structure and cost

Controlled WikiText-2 experiments use frozen GPT-2 Small, consecutive 128-token causal-language-model blocks, and one adapter beside zero-based block 0 `attn.c_proj`. Frozen projection and adapter outputs are summed before attention residual dropout and residual addition. Runs use batch size 1, gradient accumulation 4, AdamW at 3×10^{-4} , gradient clipping 1.0, no scheduler, ranks 1/2/4/8, $\alpha = 4$, and seeds 42/123/456. The independent 1,000-step screening and 5,000-step confirmation are not concatenated.

Best recorded validation, final validation at the declared step, and held-out test evaluation of the final checkpoint are distinct. Test data do not select the method, rank, hyperparameters, or checkpoint.

AG News uses the same frozen backbone and placement with a trainable classification head, 4,096 training examples, a fixed 1,000-example stratified validation subset, the original test split, 500 optimizer steps, paired seeds, and 9,216 trainable parameters including the head. Matching backbone, placement, rank, budget, optimizer, duration, and seed set limits obvious confounds; separate scale, convergence, and downstream analyses address the remaining questions.

5 Results

Table 2: Mean final-checkpoint loss $\pm$ sample SD over three seeds.

Rank	LoRA val	Symmetric val	LoRA test	Symmetric test
1	3.5734 $\pm$.0051	3.5314 $\pm$.0207	3.9256 $\pm$.0009	3.9131 $\pm$.0042
2	3.4773 $\pm$.0057	3.4340 $\pm$.0249	3.8492 $\pm$.0025	3.8305 $\pm$.0144
4	3.3483 $\pm$.0093	3.3101 $\pm$.0011	3.7656 $\pm$.0054	3.7429 $\pm$.0024
8	3.3054 $\pm$.0117	3.2446 $\pm$.0096	3.7319 $\pm$.0035	3.6938 $\pm$.0044

Paired Symmetric-minus-LoRA held-out differences are -0.0125 ± 0.0051 , -0.0187 ± 0.0119 , -0.0227 ± 0.0035 , and -0.0381 ± 0.0072 at ranks 1, 2, 4, and 8. All twelve paired seed comparisons

favour Symmetric under this protocol.

Table 3: AG News held-out results, mean $\pm$ sample SD over three seeds.

Method	Accuracy	Macro-F1	Test loss
LoRA	0.8201 $\pm$ 0.0051	0.8156 $\pm$ 0.0050	0.5445 $\pm$ 0.0542
Symmetric	0.8021 $\pm$ 0.0240	0.7929 $\pm$ 0.0315	0.6847 $\pm$ 0.1919

AG News favours LoRA on average, so the language-model ordering is task dependent.

6 Scaling, derivatives, learned spectra, and cost

Table 4: Seed-42 scaling ablation after 1,000 steps. Δ is Symmetric minus LoRA validation loss.

Rank	$\alpha = 4$			$\alpha/r = 1$		
	LoRA	Sym.	Δ	LoRA	Sym.	Δ
1	3.6171	3.6614	+0.0443	3.6301	3.6711	+0.0410
2	3.5861	3.5561	-0.0300	3.5984	3.5693	-0.0292
4	3.5530	3.4780	-0.0750	3.5530	3.4780	-0.0750
8	3.5492	3.4315	-0.1177	3.5056	3.4081	-0.0975

The single-seed constant- α/r ablation preserves the screening sign pattern but changes its magnitude; it is not generalized beyond seed 42.

Table 5: Mean paired validation difference (Symmetric minus LoRA) within fresh 5,000-step trajectories.

Rank	1,000	2,000	3,000	4,000	5,000
1	+0.0280	+0.0096	-0.0043	-0.0265	-0.0419
2	-0.0367	-0.0432	-0.0319	-0.0311	-0.0434
4	-0.0707	-0.0631	-0.0803	-0.0635	-0.0382
8	-0.1247	-0.1156	-0.0947	-0.0781	-0.0608

Rank 1 crosses between the recorded 2,000- and 3,000-step evaluations. The independent 1,000-step screen is not joined to these confirmation trajectories. The following derivative analyses characterize trained adapters; they do not by themselves explain the observed performance differences.

6.1 Local Jacobians

LoRA and Symmetric have

$$J_{\text{LoRA}} = \frac{\alpha}{r} AB, \quad (4)$$

$$J_{\text{Sym}}(x) = 2\frac{\alpha}{r} U \text{diag}(xU)P. \quad (5)$$

LoRA is linear at the adapter level, so its Jacobian is input independent and can be represented by one fixed low-rank update matrix. Symmetric is quadratic; its local transformation changes with $\text{diag}(xU)$ and cannot in general be reduced to one constant ΔW . The analytical expressions were checked against PyTorch autograd in double precision.

6.2 Effective-rank analysis

Table 6: Mean entropy effective ranks. The columns describe different mathematical objects.

Rank	LoRA update	Sym. Jacobian	Sym. U	Sym. P	Output cov.
1	1.00	1.00	1.00	1.00	1.00
2	1.92	1.67	2.00	1.98	1.92
4	3.76	3.09	3.99	3.89	3.00
8	6.57	5.59	7.96	7.28	3.59

These quantities are not interchangeable notions of “weight rank.” In particular, the input-dependent Symmetric local-Jacobian rank is not a direct analogue of a fixed LoRA weight-update rank.

6.3 Second-order structure

For fixed linear output reduction c ,

$$H = 2\frac{\alpha}{r}U \text{diag}(Pc)U^T. \quad (6)$$

LoRA has an exactly zero adapter-only Hessian because its branch is linear. Symmetric mean Hessian Frobenius norms are 155.15, 92.13, 52.68, and 17.61 at ranks 1, 2, 4, and 8. The decrease combines learned-factor differences with α/r scaling and is not evidence of weaker or less useful interactions at higher rank. For fixed c , the Hessian is input independent, so repeated token activations are not independent Hessian observations. Double-precision autograd tests verify the formula. A full Transformer-block Hessian was not evaluated and is not reported as an empirical result.

6.4 Computational cost at rank 4

The matched benchmark uses batch size 1, sequence length 128, ten warm-up iterations, and 30 measured iterations. Both adapters have 6,144 trainable parameters. Relative to LoRA, Symmetric

Table 7: Matched rank-4 computational benchmark.

Method	Forward (ms)	Backward (ms)	VRAM (MiB)	Tokens/s
LoRA	12.157 ± 0.243	14.805 ± 0.236	429.05	10,529
Symmetric	12.184 ± 0.236	14.766 ± 0.188	430.31	10,505

changes forward latency by approximately +0.22%, backward latency by −0.27%, peak VRAM by +1.25 MiB, and throughput by approximately −0.22%. The measured costs are nearly identical in this one implementation and hardware configuration; the benchmark is not generalized to other hardware or model scales.

7 Conclusion

This study investigated whether a simple quadratic transformation in a low-dimensional activation space can provide a useful alternative to conventional linear low-rank adaptation. The results show

that the Symmetric Quadratic Adapter is a practical and computationally lightweight mechanism for introducing explicit second-order interactions into a frozen Transformer while retaining the same low-rank parameter structure used for comparison with LoRA.

The strongest evidence comes from the controlled WikiText-2 experiments. Across 24 independent 5,000-step runs covering ranks 1, 2, 4, and 8 and three random seeds, Symmetric achieved lower mean final validation loss and lower mean held-out test loss than LoRA at every tested rank. The largest final-test difference was at rank 8, where mean loss decreased from 3.7319 ± 0.0035 for LoRA to 3.6938 ± 0.0044 for Symmetric. The rank-1 trajectory also depends on optimization time: LoRA initially performs better, while Symmetric overtakes it between the recorded 2,000- and 3,000-step checkpoints. Rank and duration therefore matter in addition to final parameter budget.

Holding α/r constant changes the magnitude of the differences but, for the tested seed, preserves the 1,000-step sign pattern: LoRA performs better at rank 1, whereas Symmetric performs better at ranks 2, 4, and 8. This pattern is not explained exclusively by the primary protocol’s changing scale, but the single-seed ablation prevents a broader conclusion.

AG News provides a counterexample to general superiority. LoRA achieved higher mean test accuracy and macro-F1. Accuracy was 0.8201 ± 0.0051 for LoRA and 0.8021 ± 0.0240 for Symmetric, with greater variability for the quadratic adapter. The combined evidence indicates a task-dependent trade-off.

The structural analyses establish that the adapters are genuinely different. LoRA is linear in its adapter input and has a fixed low-rank update. Symmetric has an input-dependent local Jacobian and explicit non-zero adapter-level second derivatives, so it cannot generally be reduced to one constant ΔW . Local-Jacobian effective rank, learned-factor rank, and output-covariance rank are distinct quantities rather than interchangeable notions of weight rank. These properties do not establish a causal explanation for task performance.

The added structure has little measured cost in the matched rank-4 benchmark: forward and backward latency are almost identical to LoRA, and peak memory increases by about 1.25 MiB. Thus, in this implementation, explicit quadratic activation interactions operate at approximately the same practical scale as the matched linear adapter.

Taken together, **low-rank quadratic activation-space adaptation is a viable design point for parameter-efficient Transformer adaptation, and under the controlled WikiText-2 protocol it consistently improves final language-model loss relative to matched LoRA.** The AG News result demonstrates that this advantage does not automatically transfer across tasks. Functional form therefore merits treatment as an experimental variable alongside rank and parameter budget.

The evidence remains limited to GPT-2 Small, restricted placements, two datasets, three seeds for the principal experiments, and one seed for the scaling ablation. It establishes neither state-of-the-art performance nor broad statistical significance, universal transfer, a causal link from second-order structure to loss, or universal mathematical novelty. Replication on larger models, additional tasks, broader placements, and independently repeated scaling configurations is required to determine how broadly the observed behavior generalizes.

References

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